Algo HW-3

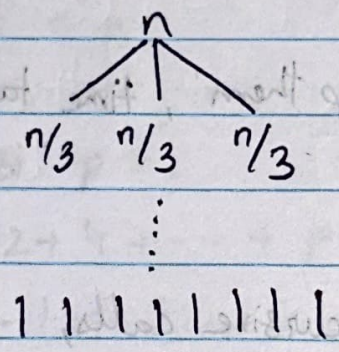
Q.1. In the line 'swap them', time taken is constant
$\therefore \Theta(1)$

There are two recursive calls;
In line 2 of code, the function is called 'n-1 times' ~~recursively~~
In line 3 of code, the function is also called 'n-1 times'

$$T(n) = \begin{cases} \Theta(1) & \text{if } n \leq 1 \\ 2T(n-1) + \Theta(1) & \text{if } n > 1 \end{cases}$$

$$T(n) = 2T(n-1) + c$$
$$= 2(2T(n-2) + c) + c$$
$$\vdots$$
$$= 2^k T(n-k) + (2^{k-1} + 2^{k-2} + \dots + 2 + 1)c$$
$$= 2^k T(n-k) + (2^k - 1)c$$
$$T(n) = 2^n T(0) + (2^n - 1)c$$
$$\therefore T(n) = \Theta(2^n)$$

Q.2.



level $i \Rightarrow \Theta(n)\ \Theta(n) \ldots\ldots \Theta(n)$

From the recursion tree,

no. of nodes at level $i = 3^i$

Size of nodes at level $i = n/3^i$

$\therefore$ Total cost $= 3^i \cdot \Theta(n/3^i) = \Theta(n)$

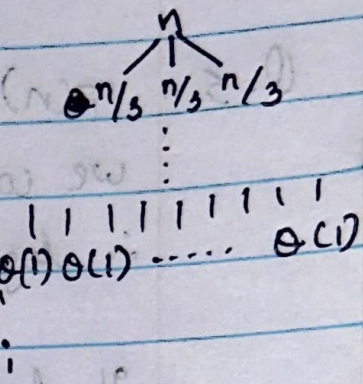
~~no. of levels $= n/3^k = 1$~~

Lets consider a constant $L = \log_3 n$ ~~such that~~, $3^L = n$

$\therefore$ no. of levels $= n/3^L = 1$

$\therefore$ Total cost $= \Theta(n) \cdot \Theta(\log n) = \Theta(n \log n)$

Q.3. $V(n) = 3V(n/3) + \Theta(1)$



$\therefore$ non-recursive cost $= \Theta(1)$

no. of nodes $= 3^i$

cost per node $= \Theta(1)$

total cost $= 3^i \cdot \Theta(1) = \Theta(3^i)$

Let height of tree be L, $\therefore n/3^L = 1$

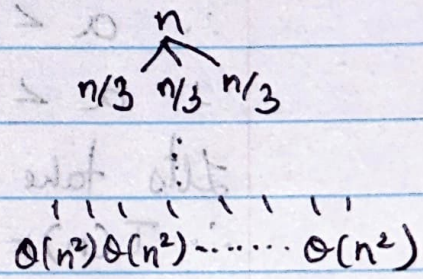
$\therefore 3^L = n$, $L = \log_3 n$

$\therefore$ total cost $= \sum_{i=0}^{L} \Theta(3^i)$ ~~$= \Theta(3^L) = \Theta(n)$~~

$= \Theta(3^L)$

$= \Theta(n)$

Q.4 $W(n) = 3W(n/3) + \Theta(n^2)$



$\therefore$ non-recursive cost $= \Theta(n^2)$

no. of nodes $= 3^i$

cost per node $= \Theta\left(\frac{n^2}{9^i}\right)$

Size of node $= n/3^i$

total cost $= 3^i \cdot \Theta(n^2/9^i) = \Theta\left(n^2 (1/3)^i\right)$

recursion stops when, $n/3^L \approx 1$, $L = \log_3 n$

$\therefore \sum_{i=0}^{L} \Theta(n^2 (1/3)^i) = \Theta\left(n^2 \sum_{i=0}^{L} (1/3)^i\right) = \Theta(n^2) \cdot \Theta(1) = \Theta(n^2)$

Q.5. $T(n) = aT(n/5) + \Theta(n)$,

we compare $n^{\log_5 a}$ to $f(n) = \Theta(n)$,

$\therefore \log_5 a = 1$

If $\log_5 a < 1$, ie. $a < 5$

then $T(n) = \Theta(n)$

If $\log_5 a = 1$, ie. $a = 5$

then $T(n) = \Theta(n \log n)$

If $1 < \log_5 a < 2$, ie. $5 < a < 25$

then $T(n) = \Theta(n^{\log_5 a})$

we need complexity strictly less than $\Theta(n^2)$

$\therefore \log_5 a < 2$

$\therefore a < 5^2$

$\therefore a < 25$

Lets take $a = 25$

$\therefore T(n) = \Theta(n^{\log_5 25})$

$= \Theta(n^2)$

Q.6 $S = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]$

pop top $i = 3$,
$\therefore S = [3, 4, 5, 6, 7, 8, 9]$

pop ~~[illegible]~~ 4 elements,
$\therefore S = [7, 8, 9]$, Since 3, 4, 5, 6 are popped.

pushing the same elements in same order,
$\therefore S = [6, 5, 4, 3, 7, 8, 9]$

pop ~~order~~ 2, 1, 0 in S respectively,
$\therefore S = [0, 1, 2, 6, 5, 4, 3, 7, 8, 9]$